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Github: https://github.com/Programlog/agentic-NBA-prediction-CMSC473.git
Proposal Slides: **NBA Shot Prediction Proposal - CMSC473**

Problem

Predicting whether an NBA shot will be successful is difficult as there are a plethora of factors that can impact a shooting motion. These include quantitative factors like shot location, defender proximity, time left in game, time left on shot clock, player statistical history, and more. However, there are also factors related to shooting motion that are not captured in traditional NBA statistics. These include the revolutions on the ball from shot, defender close out height as well as shot taker angle, and we would potentially have to extract these factors from video footage.

Our project will be investigating two research questions. One question is whether computer-vision derived features can help predict NBA shot success better than only relying on traditional NBA statistics. Our second question is that when given the same information and features, will a multi-agent LLM pipeline contribute more towards NBA shot success prediction than a single LLM baseline. By separating the Computer Vision and Agentic AI aspects, we will be able to see whether we see larger improvements from using additional computer vision features, the agentic AI architecture, or both.

We will evaluate whether the addition of features derived from computer vision improves our baseline prediction models such as XGBoost. We also will evaluate whether a multi-agentic AI pipeline will perform better in shot success prediction in comparison to XGBoost, a single LLM baseline, when using controlled inputs.

Our work done in this project can be useful for player development, specifically for evaluating shot quality, defensive performance, and player decision-making. Additionally, our novel approach merges NBA statistics with computer-vision derived features and an agentic AI system while evaluating the importance of each component individually. Since shot outcomes are random by nature, we are not aiming to perfectly predict shot success, however we are looking to produce better-calibrated shot success probabilities.

Literature Review:

Predicting the success of an NBA shot has been done in two ways: expected value approaches based on player tracking data and standard classification models using publicly available shot logs. Chang et al. (2014) applied SportVU tracking data to develop Effective Shot Quality, an expected value of each shot considering location and defender proximity, where quality of a shot and its success were dissociated. Cervone et al. (2016) extended the approach by proposing Expected Possession Value, a stochastic model assigning point value to every state of possession during play and including probability of shot make as one of its components. They proved that location and defender distance are the most important public information, but do not include anything about mechanics of shooting.

The proof of the importance of mechanics of shooting is provided by the proprietary sensor system. Daly-Grafstein and Bornn (2019) demonstrated that conditioning shot estimation on ball entry angle and depth gives an estimator with much lower variance than regular field goal percentage and is a better predictor of shooting in future seasons. Such data can be obtained only by proprietary systems like Noah and Hawk-Eye.

The performance when only using publicly available shot-log data seems to be somewhat limited. Meehan (2017), using the Kaggle 2014-15 shot logs (about 128,000 shots with distance, closest defender distance, touch time, and shot clock), finds logistic regression with 59% accuracy, random forest with 61%, and XGBoost with 68%, while other student re-runs cluster at about 65%. Prior results when using pre-release positional features are around the mid-60s based on 45% make rate. Grinsztajn et al. (2022) found gradient boosted trees are still the best learners for tabular data at medium scale, so a properly tuned XGBoost is the natural baseline rather than a straw man.

Mechanics from broadcast video are possible for some features. RTMPose (Jiang et al., 2023) offers real-time multi-person pose estimation with permissive licensing, and Nakai et al. (2018) have shown that pose features are sufficient to predict free-throws from video. Contest distance needs a homography of the court, and backspin on the ball cannot be extracted from broadcast video.

Lastly, the value that multi-agent LLM pipelines provide is still unknown. Halawi et al. (2024) developed a pipeline of "retrieve-reason-aggregate" that rivals human crowd performance, but the majority of improvement was attributed to the retrieval of fresh knowledge rather than to the orchestration of agents, and the probabilities provided by LLMs are known to be overconfident. This gives us motivation to test whether agentic AI architecture can add value beyond a single LLM baseline when using the same inputs.

As a whole, no publicly available project integrates vision-driven shooting mechanics along with a tabulated shot log. Furthermore, no research has explored whether LLM agents could outperform a calibrated XGBoost baseline in predicting shots.

Why can't AI do this end to end?

1. Need to acquire both nba statistics and real time footage
2. Choose and implement a computer vision model and optimize
3. Synchronize video data with real time data
4. Need to define what counts as a release angle, contest distance etc
5. Deciding whether detected player is defender or shooter
6. Designing non leaking experiments
7. Handle and interpret surprising results

Research scope

- Focus on predicting the likelihood that an NBA shot attempt is made or missed.
- Use traditional shot data like distance, defender proximity, context of game, statistical history, and more.
- Use video footage to extract additional computer-vision features like release angle, defender closeout, and player positioning.
- Evaluate the addition of computer-vision derived features by analyzing XGBoost with and without the additional features.
- Evaluate the contribution of the agentic AI architecture by comparing the multi-agent pipeline with the single-LLM baseline and XGBoost models with the same inputs.

Datasets (find or create)

- Kaggle NBA shots (https://www.kaggle.com/datasets/dansbecker/nba-shot-logs)
    - NBA Shot Logs dataset by Dan Becker
    - Contains roughly 128,000 shot attempts from the 2014-2015 NBA regular season.
- Official nba_api
    - NBA.com stats and NBA live data.
    - Since it is an API and not a static dataset, there isn't a fixed size.
- Video source VideoEventsAsset
    - Endpoint in the NBA Stats API to retrieve videos.
    - There isn't a fixed size because videos are retrieved using the shots/games that you query.
- sealneaward/nba-movement-data github
    - Github repo containing archived NBA player/ball movement data by neilmj (the NBA stopped providing this type of data).

    - Stores data from approximately 636 NBA games
- linouk23/NBA-Player-Movements github
    - Github repo containing SportVU game logs from a sample of 2015-2016 NBA regular season games
    - Stores data from approximately 600 NBA games.

Models (specify existing modes to build or evaluate):

- Baseline prediction (Logistic Regression)
- Main tabular model (XGBoost)
- Player and ball detection/ tracking (latest YOLO model)
- Body pose (YOLO pose)
- Multi-Agentic Pipeline (GPT-6 Luna)
- Single-Agent Baseline (GPT-6 Luna)

Compute:

- We will utilize local Machines + Google Colab for preprocessing, baseline ML model training, and agent development
- We will need a CUDA GPU for YOLO/YOLO-Pose inference on video frames
- Primarily CPUs for Baseline Models
- Multi-agent pipeline using CrewAI will be using either open-source LLMs or API-hosted LLMs
- We expect that we won't require multi-GPU or multi-node computing for the current scope

Approach

- Use Python and Pandas to build a data processing pipeline to clean and combine NBA shot logs, player statistics, tracking data, and also matching video clips.
- Train our baseline Logistic Regression and XGBoost models using traditional features like shot distance, defender proximity, game clock, shot clock, player statistics, and more.
- Use YOLO or YOLO-Pose with OpenCV to identify players and the ball, which can help calculate features such as shooter release angle, defender positioning, and closeout height.
- Train a XGBoost mode that now uses both the traditional features as well as the features derived from the Computer Vision process.
- Build a multi-agent system using CrewAI (will assign roles to our agents and pass outputs from one agent to another to provide more context, and also feed the results to a prediction agent), where specialized agents analyze player history, shot-mechanics, game context from different perspectives. Using their outputs a prediction agent will estimate the likelihood of a shot being made.
- Evaluate all the models on held-out data (games) using Brier score, log loss, accuracy and ROC-AUC.

Validation:

- Unit tests will be made to run throughout different milestones in our project to validate data preprocessing, feature extraction, and model outputs.
- To ensure that our computer vision component is providing accurate results, we will take a sample of our videos and check that the data pertaining to the features that were extracted from the videos look reasonable and match the actions occurring in the videos.
- We will test our models on the same data (games) that were not used during training, so that we can analyze how our system works with unseen examples, which can help us make any adjustments if needed.
- We will compare model performance using measurements like accuracy as well as probability calibration (Brier score), log loss, ROC-AUC. Emphasis will be placed on probability calibration

and analyzing any failure cases.

Deliverables:

- Analysis using ML Model (Baseline model): After cleaning a NBA shot dataset, we employ logistic regression and XGBoost to predict shot success.
- Computer Vision analysis: Extracting additional factors relating to shooting from NBA video clips , such as release angle, defender position, contest height using object detection.
- Computer Vision Enhanced Model: Train XGBoost using features from traditional shot data as well as computer vision derived features.
- Agentic AI System: A multi-agent system consisting of a Context Agent, Player History Agent, Shot Mechanics Agent, Defender context Agent and more specialized agents that feeds a prediction agent to output the probability of a shot being successful
- Single-Agent baseline: Use one Agent with the same available inputs to compare with the multi-agent system.
- Evaluation: Comparison of our baseline XGBoost, CV enhanced XGBoost, single LLM system and our agentic AI system. This comparison will utilize Brier-score, log loss, accuracy, and ROC-AUC.

Timeline & Milestones

1. Weeks 1–2
    a. Collect shot tracking logs from the NBA API.
    b. Download matching short video clips for a selected sample of shots.
    c. Set up code repositories and compute environments.
2. Weeks 3–4
    a. Train baseline ML models (Logistic Regression, XGBoost) on tabular data.
    b. Build a computer vision script (using tools like YOLO and MediaPipe) to extract player release angle and defender contest height from video clips.
3. Weeks 5–6
    a. Build the multi-agent system: Context Agent, Player History Agent, and Shot Mechanics Agent.
    b. Connect the agents to the data pipeline and prompt the aggregator agent to produce the final probability.
4. Weeks 7–8
    a. Compare XGBoost with and without CV, multi-agent system vs single LLM vs CV-enhanced XGBoost.
    b. Refine agent prompts and vision feature accuracy.
5. Weeks 9–10
    a. Analyze failure cases (shots the model got completely wrong)
    b. Prepare the final report, slides, and demo.

References

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